THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student wants to find out how oil temperature affects the puffing of a puri. The factor she deliberately changes is called the: 1

(a) independent variable **(b)** dependent variable
(c) conclusion **(d)** constant

2. Who first observed and named 'cells' in thin slices of cork in 1665? 1

(a) Robert Hooke **(b)** Louis Pasteur
(c) Anton van Leeuwenhoek **(d)** Robert Brown

3. According to the World Health Organization, health is a state of complete: 1

(a) physical strength only **(b)** physical, mental and social well-being
(c) wealth and comfort **(d)** absence of fever only

4. A device that provides the electrical energy to drive a current in a circuit is a: 1

(a) wire **(b)** cell or battery **(c)** switch **(d)** bulb

5. A force is a: 1

(a) type of energy only **(b)** state of rest
(c) kind of speed **(d)** push or a pull

6. Pressure is defined as: 1

(a) force acting per unit area **(b)** force multiplied by area
(c) mass per unit volume **(d)** area per unit force

7. According to the particulate model, matter is made up of: 1

(a) tiny particles **(b)** only air
(c) continuous jelly-like material **(d)** only liquids

8. A pure substance that cannot be broken down into simpler substances by chemical methods is a(n): 1

(a) element **(b)** solution **(c)** compound **(d)** mixture

9. A homogeneous mixture of two or more substances is called a: 1

(a) element **(b)** solution **(c)** suspension **(d)** compound

10. According to the first law of reflection, the incident ray, the reflected ray and the normal at the point of incidence: 1

(a) are parallel **(b)** are always equal in length
(c) lie in the same plane **(d)** are at right angles to each other

11. The apparent daily motion of the Sun from east to west is caused by the: 1

(a) revolution of the Earth around the Sun **(b)** rotation of the Earth on its axis
(c) rotation of the Sun **(d)** motion of the Moon

12. An ecosystem consists of: 1

(a) only soil and water
(b) only plants and animals
(c) only humans
(d) living and non-living components interacting with each other

13. The Earth is the only known planet that supports life because it has: 1

(a) rings **(b)** only a Moon
(c) water, air and a suitable temperature **(d)** no atmosphere

14. A scientific investigation usually begins with: 1

(a) writing a final report
(b) buying equipment
(c) a question or curiosity about an observation
(d) announcing the result

15. The correct order of increasing complexity is: 1

(a) cell, tissue, organ, organ system, organism
(b) organ, tissue, cell, organism
(c) tissue, cell, organ, organism
(d) cell, organ, tissue, organism

16. A sign of a disease is: 1

(a) only what the patient feels
(b) a medicine
(c) something a doctor can observe or measure, such as a rash
(d) a food habit

17. The heating element of an electric heater is made of: 1

(a) copper **(b)** iron **(c)** plastic **(d)** nichrome

18. Friction is greater between: 1

(a) oiled surfaces **(b)** smooth surfaces
(c) polished surfaces **(d)** rough surfaces

19. The instrument used to measure atmospheric pressure is a: 1

(a) thermometer **(b)** barometer **(c)** speedometer **(d)** hygrometer

20. The change from solid to liquid is called: 1

(a) condensation **(b)** boiling **(c)** melting **(d)** freezing

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. What is a hypothesis? Give one example related to puri making. 2

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22. Who discovered cells and how? Name the material he observed. 2

23. Define health as per the World Health Organization. 2

24. What is an electric circuit? 2

25. What is a force? State its SI unit. 2

26. Define pressure and write its formula and SI unit. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Describe how you would plan an investigation to find the effect of oil temperature on the time taken by a puri to puff up. 3

28. Describe how you would prepare a slide of an onion peel and observe it. 3

29. Explain the three aspects of health with one example of each. 3

30. Draw in words a circuit with a cell, bulb and switch and explain how the bulb glows. 3

31. Explain how force can change the shape of an object. Give two examples. 3

OR

A box of weight 600 N rests on a floor with a base of area 0.5 square metre. Find the pressure. 3

32. Describe an activity to show that particles of matter have spaces between them. 3

OR

List three physical properties of metals with an example for each. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Design a complete investigation to study whether the amount of water used in kneading dough affects how a puri puffs. Mention question, hypothesis, variables, constants, procedure and recording. 5

34. Draw in words and label the parts of a typical plant cell and state the function of each. 5

35. Explain the concept of health as defined by WHO. Describe the physical, mental and social aspects with examples. 5

36. Describe Oersted's experiment with a labelled set-up and give its conclusion. State the effect of reversing the current. 5

OR

Describe the effects of force with five examples. 5

37. Explain, with examples, how pressure depends on force and area. Calculate the pressure of a 50 N force over 0.25 square metre and over 0.5 square metre. 5

OR

Explain the particulate nature of matter with the help of three activities. 5

38. Compare elements, compounds and mixtures with examples in a table of five points. 5

OR

Describe the characteristics of solutions, suspensions and colloids in a table of five points with examples. 5

END OF QUESTION PAPER **Total: 80 Marks**

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